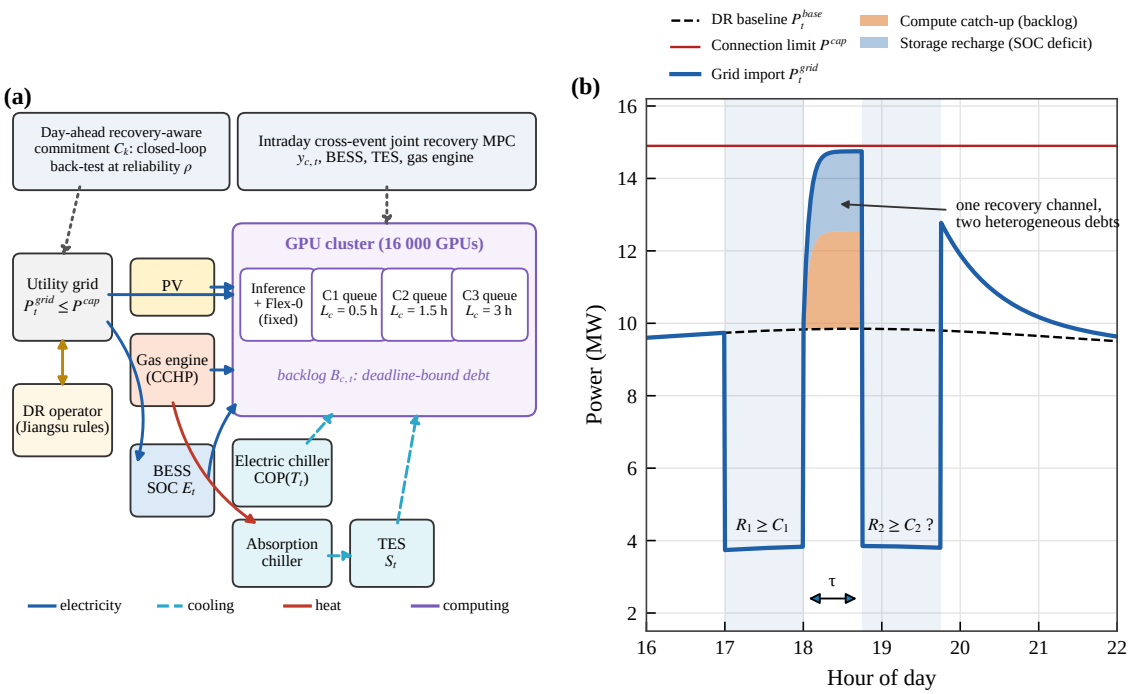
(a)
Day-ahead recovery-aware commitment C_t: closed-loop back-test at reliability ρ
Intraday cross-event joint recovery MPC $y_{c,t}$, BESS, TES, gas engine
GPU cluster (16 000 GPUs)
Inference + Flex-0 (fixed)
C1 queue L_c = 0.5 h
C2 queue L_c = 1.5 h
C3 queue L_c = 3 h
backlog $B_{c,t}$: deadline-bound debt
Utility grid $P_t^{grid} \le P^{cap}$
PV
Gas engine (CCHP)
DR operator (Jiangsu rules)
BESS SOC E_t
Electric chiller COP(T_t)
Absorption chiller
TES S_t
electricity
cooling
heat
computing
(b)
DR baseline P_t^{base}
Connection limit P^{cap}
Grid import P_t^{grid}
Compute catch-up (backlog)
Storage recharge (SOC deficit)
one recovery channel, two heterogeneous debts
$R_1 \ge C_1$
$R_2 \ge C_2$?
τ
Power (MW)
Hour of day
16
14
12
10
8
6
4
2
16
17
18
19
20
21
22